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A. Notation and Symbols

This section fixes notation and collects basic identities used throughout the appendix.

Table 1. Key notation used in the analysis.

Notation	Description
Concepts and datasets	
$\mathcal{C} = \mathcal{C}_1 \times \cdots \times \mathcal{C}_k$	Concept space with $\lvert\mathcal{C}_i\rvert = n$
$\mathcal{X} = \{\boldsymbol{x_c} \mid \boldsymbol{c} \in \mathcal{C}\}$	Input space
$\mathcal{Z} = \{\boldsymbol{z_c} \mid \boldsymbol{c} \in \mathcal{C}\}$	Representation space
$\mathcal{D}^{\boldsymbol{c}}$	Cross-dataset of size $1 + k(n-1)$ (see Definition 6)
Models and learning	
T	Training support (a subset of $\mathcal{C}$)
$\mathcal{T}$	Validity class (set of valid supports)
R	Validity rule $R : 2^{\mathcal{C}} \to \{0, 1\}$
D_T	Training dataset supported on T
$f : \mathcal{X} \to \mathcal{Z}$	Encoder / representation map
$g : \mathcal{Y} \to \mathcal{Z}$	Text encoder (CLIP instantiation)
h_T	Readout learned from D_T
$\mathcal{H}$	Readout hypothesis class
A	Learning rule mapping $D_T \mapsto h_T$
$\Pi = (f, \mathcal{H}, A, \mathcal{T})$	Compositional-generalization setup
Counts	
$N_{i,j}(S)$	Marginal count of concept i taking value j in dataset S
Interventions	
$\boldsymbol{c}(i \to j)$	Concept index with the i-th value set to j
$\boldsymbol{z}_{\boldsymbol{c}(i\to j)}$	Intervened representation with concept i set to j
$\bar{c}_i$	Binary complement $1 - c_i$ (when $\mathcal{C}_i = \{0, 1\}$)
Probes and parameters	
$\boldsymbol{w}_{i,j}$	Probe vector for concept i, value j
$\boldsymbol{W}, \boldsymbol{b}$	Linear readout parameters $h(\boldsymbol{z}) = \boldsymbol{W}\boldsymbol{z} + \boldsymbol{b}$
$\boldsymbol{w}_{i,j}^{(\mathcal{D}^{\boldsymbol{c}})}$	Weight vector for concept i, class j
$b_{i,j}^{(\mathcal{D}^{\boldsymbol{c}})}$	Bias term for concept i, class j
$\mathcal{R}_{i,j}(h)$	Region where h predicts value j for concept i
$p_i^{(T)}(\cdot \mid \boldsymbol{z})$	Per-concept posterior under readout trained on T
Factorization objects	
$\boldsymbol{P} \in \mathbb{R}^{d\times d}$	Projection matrix
$\boldsymbol{P_W}$	Orthogonal projector onto probe span
$\boldsymbol{u}_{i,c_i} \in \mathbb{R}^d$	Linear factor for concept i, value c_i